4. Find which of the sequences given below are monotonic:

(a) $y_n = 3n^2 - n$ $\quad y_{n+1} = 3(n+1)^2 - (n+1)$ $\quad y_{n+1} - y_n = 3(n+1)^2 - (n+1) - 3n^2 + n$

$= 6n + 2 = \boxed{2(3n+1)}$

$3n+1 > 0 \Rightarrow \boxed{n > -\frac{1}{3}} \Rightarrow \forall n \in N.$

The value of difference for $n = x$ and $x+1$ are respectively $2(3x+1)$ and $2(3x+4)$. This holds true for $\forall x \in N$.

$\therefore$ The sequence is increasing. ✓

(b) $y_n = n^2 - 3n$ $\quad y_{n+1} = (n+1)^2 - 3(n+1)$ $\quad y_{n+1} - y_n = (2n+1) - 3n = \boxed{1-n}$

$1 - n > 0 \Rightarrow \boxed{1 > n}$

$n = x$. Difference $= 1 - x$

$n = x+1$ Difference $= 1 - (x+1) = -x$

$\forall n > 1$, the difference is negative

$\forall x \in N, 1 - x > -x$

(Wrong)

(b) $y_n = n^2 - 3n$ $\quad y_{n+1} = (n+1)^2 - 3(n+1)$ $\quad y_{n+1} - y_n = (2n+1) - 3 = 2(n-1)$

$\forall n > 1$, the difference is +ve

$n = x$. Difference $= 2(x-1)$

$n = x+1$. Difference $= 2(x+1-1) = 2x$ $\quad \boxed{2x > 2(x-1)} \rightarrow \forall x \in N.$

(Monotonic, increasing) ✓

(c) $y_n = 7n - n^2$ $\quad y_{n+1} = 7(n+1) - (n+1)^2$ $\quad y_{n+1} - y_n = 7 - 2n - 1 = 6 - 2n = \boxed{2(3-n)}$

$n = x$. Difference $= 2(3-x)$

$n = x+1$. Difference $= 2(3-x-1) = 2(2-x)$

Upto $n = 3$, the difference is positive, then the difference is negative.

$\therefore$ Non-monotonic ✓

(d) $y_n = \log\left(\frac{3}{4}\right)^n$ $\quad y_{n+1} = \log\left(\frac{3}{4}\right)^{n+1}$

$y_{n+1} - y_n = \log\left(\frac{3}{4}\right)^{n+1} - \log\left(\frac{3}{4}\right)^n = (n+1)\log\left(\frac{3}{4}\right) - n\log\left(\frac{3}{4}\right)$

$= \log\left(\frac{3}{4}\right) = -0.124$

$\therefore$ Decreasing sequence ✓